

Assignment III

1. Finding Eigenvectors by Hand

This is a tedious exercise. But I want you to go through to practice the skill of finding eigenvectors. We can also measure the electron spin in $\hat{y}$ direction by setting up an external magnetic field in the $\hat{y}$ direction. It is given that the corresponding operator is $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. Again, we are using the $|0\rangle/|1\rangle$ basis. Find the eigenvalues and eigenvectors in the $|0\rangle/|1\rangle$ basis.

2. Finding Eigenvectors Using Python

Use Google Colab to check your answer. What you need is the function to find the eigenvectors and eigenvalues. The following code first builds the matrix and then find the eigenvectors and eigenvalues:

```
import numpy as np
sigmay=np.array([[0,-1j],[1j,0]])
print(sigmay)
eigenvalues, eigenvectors = np.linalg.eig(sigmay)
print(eigenvalues) print(eigenvectors)
```

3. Pauli Matrix Multiplication by Hand and Using Python

Use hand calculation to show that $\sigma_y^2 = I$. Then verify with Colab using the following codes. Note that np.matmul is for matrix multiplication.

```
import numpy as np
sigmay=np.array([[0,-1j],[1j,0]])
print(sigmay)
print(np.matmul(sigmay, sigmay))
```

4. Pauli Matrix Multiplication by Hand and Using Python

Repeat the previous question for $\sigma_z^2 = I$ using both hand calculation and Colab.

5. Proof of Hermitianity

Show that σ_y is Hermitian. Hints: find σ_y^{*T} and show that it is the same as σ_y . Verify using Colab.

6. Operation of Paul Matrix on Bra and Ket

Find $\sigma_y|0\rangle$, $\langle 0|\sigma_y$. Are they the same? Why? Verify using Colab. You can use np.dot for matrix-vector multiplication. The following shows how to do $\sigma_y|0\rangle$.

```
zerospin=np.array([[1],[0]])
print(zerospin.shape)
np.dot(sigmay,zerospin)
```